

SAT Math Mastery · Week 2B

Linear Modeling & Inequality Regions

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics thresholds and tiered rates · mixtures · whole-number constraints · linear inequalities in context · shaded regions and inequality systems · feasible points · triangle inequality · unit shifts · meaning of parameters

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Build a linear function with a flat part, a threshold and a rate: $C = F + r(n - k)$ for $n \geq k$.
- S2.** Handle tiers, discounts, tax and percent factors in the correct order.
- S3.** Translate phrases (at most, at least, between, no more than) into inequalities and solve them.
- S4.** Round whole-number answers in the feasible direction, and check the boundary value.
- S5.** Model two-item mixtures and percent mixtures with a count equation and a value equation.
- S6.** Read a shaded region: find the boundary line, decide solid or dashed, test a point for the direction.
- S7.** Decide whether an ordered pair satisfies a system of inequalities (every inequality, strict vs. non-strict).
- S8.** Use the triangle inequality, unit shifts through a linear conversion, and the meaning of slope and constants in context.

2 · REQUIRED FACTS

- “At most” means $\leq$; “at least” means $\geq$; “fewer than” means $<$; “between a and b , inclusive” means $a \leq x \leq b$.
- Multiplying or dividing an inequality by a negative number reverses the sign.
- A solid boundary line includes its points ($\leq, \geq$); a dashed boundary line does not ($<, >$).
- A point is a solution of a system of inequalities only if it satisfies *every* inequality.
- The graph of $y \geq mx + b$ is shaded above the line, $y \leq mx + b$ below it; always confirm with a test point such as $(0, 0)$ when it is not on the line.
- For whole-number quantities (items, hours, people) the answer is an integer on the feasible side of the boundary.
- Triangle inequality: each side is shorter than the sum of the other two; the third side x satisfies $|a - b| < x < a + b$.
- A change Δx in the input of $y = mx + b$ changes the output by $m \Delta x$; the constant b cancels.

3 · REQUIRED FORMULAS

$C(n) = F + r(n - k)$ ($n \geq k$), total with tax rate p : $(1 + p) \cdot \text{pre-tax total}$, after $p\%$ off: $(1 - p) \cdot \text{price}$

mixture: $x + y = T$, $c_1x + c_2y = cT$, $|a - b| < x < a + b$, $\Delta y = m \Delta x$

feasible region: all inequalities hold together; corner points come from intersecting two boundary lines

4 · MINI THEORY SHEET

Phrase to inequality.

Words	Symbol and round-off rule
at most, no more than, cannot exceed	$\leq$; for a maximum count, round <i>down</i>
at least, no fewer than, minimum of	$\geq$; for a minimum count, round <i>up</i>
more than, exceeds, less than	$>$ or $<$; test the boundary value itself
between a and b	$a < x < b$ (inclusive: $a \leq x \leq b$)

Thresholds and tiers. Write the cost of each tier separately and add them: first k units at the flat price, then rate $\times$ (units $- k$). Apply a discount only to the units it covers; apply tax to the final subtotal.

Mixtures. Use a count equation and a value equation. For concentrations: (amount) $\times$ (concentration) summed over the parts equals (total amount) $\times$ (final concentration). Adding pure water or a pure substance means that part has concentration 0% or 100%.

Regions. Find two lattice points on the boundary, write the line, then test a point not on it. Solid line $\Rightarrow \leq$ or $\geq$; dashed line $\Rightarrow <$ or $>$. For an overlap region, repeat for each boundary.

Feasible points. Test a candidate in every constraint. The correct choice satisfies all of them; each wrong choice usually breaks exactly one.

Meaning in context. In $y = mx + b$: m is the change in y per 1 unit of x , b the value at $x = 0$. A product such as $0.85 \cdot 6$ combines both a discount factor and a rate: ask “per what?”.

Speed shortcuts.

Situation	Shortcut
maximum whole-number count	solve the inequality, then take the floor and recheck the cost
extra units beyond a threshold	charge only $(n - k)$ units at the new rate
percent change and tax together	multiply the factors: 0.65×1.08
two-item mixture, one unknown asked	total value $-$ (all cheaper item) $\div$ price gap
shaded region	line through two lattice points, then test the origin
integer points in a system	plug in the endpoints of each inequality, strict bounds excluded
triangle with sides a, b, x	$ a - b < x < a + b$; count integers carefully
degrees F vs. C (or K)	$\Delta F = 1.8 \Delta C$; ignore the $+32$

5 · GUIDED EXAMPLES

Example 1 — Threshold and whole-number rounding

A taxi charges \$4 for the first 2 miles and \$1.80 for each additional mile (whole miles only). What is the greatest trip, in miles, that costs no more than \$20?

For $m \geq 2$: $C = 4 + 1.80(m - 2) \leq 20 \Rightarrow m - 2 \leq \frac{16}{1.8} \approx 8.89$. Whole miles: round *down* to $m - 2 = 8$, so $m = 10$. Check: $m = 10$ costs \$18.40; $m = 11$ costs \$20.20 $>$ \$20.

Example 2 — From a shaded region to an inequality

The boundary is a dashed line through $(0, 4)$ and $(2, 0)$; the side containing the origin is shaded. Write the inequality.

Slope $= \frac{0 - 4}{2 - 0} = -2$, so $y = -2x + 4$, that is $2x + y = 4$. Test $(0, 0)$: $0 < 4$ is true, so the shaded side is $2x + y < 4$ (dashed line, so strict).

Example 3 — Percent mixture

How many liters of a 10% solution and of a 40% solution make 30 liters of a 20% solution?

$x + y = 30$ and $0.10x + 0.40y = 0.20(30) = 6$. Substitute $x = 30 - y$: $3 + 0.30y = 6 \Rightarrow y = 10$ liters of the 40% solution and $x = 20$ liters of the 10% solution.

6 · PRACTICE SET — 37 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

A streaming service charges a one-time sign-up fee of \$12 plus \$9 per month. Priya wants to spend no more than \$100 in total. Which inequality can be used to find m , the number of months she can subscribe?

(A) $9(12 + m) \leq 100$

(B) $12 + 9m \leq 100$

(C) $12m + 9 \leq 100$

(D) $12 + 9m \geq 100$

2. EASY · NO CALCULATOR NEEDED

Two sides of a triangle have lengths 7 and 12. Which of the following could be the length of the third side?

(A) 21

(B) 5

(C) 11

(D) 19

3. EASY · NO CALCULATOR NEEDED

The temperature in degrees Fahrenheit, F , is related to the temperature in degrees Celsius, C , by $F = 1.8C + 32$. If the temperature rises by 10 degrees Celsius, by how many degrees Fahrenheit does it rise?

4.

EASY · NO CALCULATOR NEEDED

$$y > 2x - 1$$

$$y < -x + 5$$

Which of the following ordered pairs (x, y) is a solution to the system of inequalities above?

(A) $(1, 2)$ (B) $(2, 3)$ (C) $(-1, 6)$ (D) $(4, -2)$

5.

EASY · CALCULATOR

A parking garage charges \$6 for the first 2 hours and \$2.50 for each additional hour. What is the cost, in dollars, of parking for 7 hours?

6.

MEDIUM · NO CALCULATOR NEEDED

A delivery service charges a flat \$8 for the first 3 kilometers and \$1.50 for each kilometer after the first 3. Which equation gives the cost C , in dollars, of a trip of d kilometers, where $d \geq 3$?

(A) $C = 1.50(d - 3)$ (B) $C = 8 + 1.50(d + 3)$ (C) $C = 8 + 1.50d$ (D) $C = 8 + 1.50(d - 3)$

7.

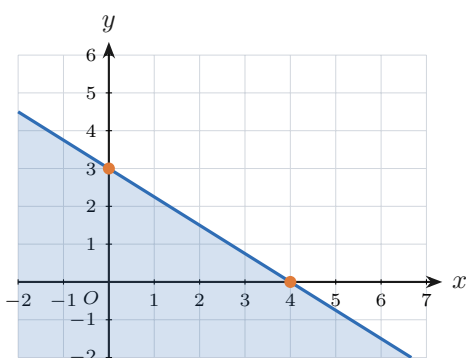
MEDIUM · CALCULATOR

An elevator can carry at most 1,200 pounds. An operator who weighs 160 pounds is in the elevator, and each box loaded weighs 85 pounds. What is the greatest number of boxes that can be loaded?

8.

MEDIUM · NO CALCULATOR NEEDED

The shaded region in the xy -plane is the graph of a linear inequality, and the boundary line is solid. Which inequality represents the shaded region?



(A) $4x + 3y \leq 12$

(B) $3x + 4y < 12$

(C) $3x + 4y \leq 12$

(D) $3x + 4y \geq 12$

9.

MEDIUM · CALCULATOR

A shop mixes coffee beans that cost \$8 per pound with coffee beans that cost \$12 per pound to make 20 pounds of a blend that costs \$9.50 per pound. How many pounds of the \$12 beans are in the blend?

10.

MEDIUM · NO CALCULATOR NEEDED

A triangle has side lengths 6, 10 and x , where x is an integer. What is the sum of all possible values of x ?

11.

MEDIUM · NO CALCULATOR NEEDED

A museum membership costs \$40 plus \$6 per visit, and members receive 15% off their total. The member's total cost, in dollars, for v visits is $C = 0.85(40 + 6v)$. Which is the best interpretation of the value $0.85 \cdot 6$ in this context?

- (A) The percent discount on each visit (B) The cost of one visit after the discount
(C) The amount saved on each visit (D) The cost of 6 visits after the discount

12.

MEDIUM · NO CALCULATOR NEEDED

Maria earns p dollars per month and saves 15% of it. She wants her monthly savings to be at least \$270 and at most \$450. Which inequality gives all possible values of p ?

- (A) $317.6 \leq p \leq 529.4$ (B) $40.5 \leq p \leq 67.5$
(C) $270 \leq p \leq 450$ (D) $1800 \leq p \leq 3000$

13.

MEDIUM · NO CALCULATOR NEEDED

A student buys n notebooks at \$2.50 each and p pens at \$1.25 each. She spends at most \$30, buys at least 4 notebooks, and buys at least twice as many pens as notebooks. Which of the following pairs (n, p) is possible?

(A) (5, 10)

(B) (3, 8)

(C) (5, 9)

(D) (6, 16)

14.

HARD · NO CALCULATOR NEEDED

A mobile data plan costs \$20 per month and includes the first 5 gigabytes (GB). Each of the next 10 GB costs \$3 per GB, and each GB beyond the first 15 GB costs \$6 per GB. Data is billed in whole GB. What is the greatest amount of data, in GB, that can be used in a month for a total cost of at most \$100?

(A) 31

(B) 23

(C) 24

(D) 28

15.

HARD · CALCULATOR

A store sells jackets for \$60 each. A customer pays full price for the first jacket, and each additional jacket is 35% off. An 8% sales tax is then added to the total. What is the greatest number of jackets a customer can buy with \$400?

- (A) 6 (B) 7
(C) 8 (D) 9

16.

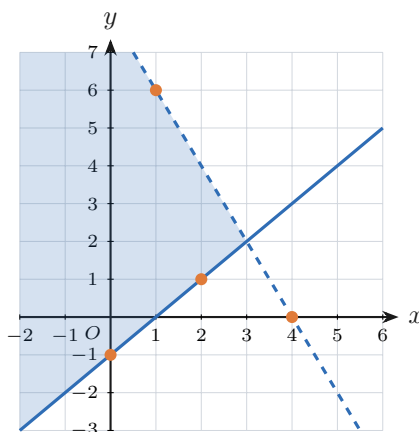
HARD · NO CALCULATOR NEEDED

A club has \$900 to spend on T-shirts and hoodies. Each T-shirt costs \$9 and each hoodie costs \$24. The club wants to buy at least 50 items in all and at least twice as many T-shirts as hoodies. What is the greatest number of hoodies the club can buy?

17.

HARD · NO CALCULATOR NEEDED

The shaded region in the xy -plane is the solution set of a system of two linear inequalities. One boundary line is solid and the other is dashed. Which system of inequalities has the shaded region as its solution set?



(A) $y \geq x - 1$ and $y < -2x + 8$

(B) $y \leq x - 1$ and $y < -2x + 8$

(C) $y \geq x - 1$ and $y > -2x + 8$

(D) $y > x - 1$ and $y \leq -2x + 8$

18.

HARD · NO CALCULATOR NEEDED

$$y > 2x - 5$$

$$2y \leq 3x + 11$$

For how many integer values of k is the point $(4, k)$ a solution to the system of inequalities above?

19.

HARD · CALCULATOR

A 240-milliliter solution is 35% alcohol by volume. How many milliliters of pure water must be added to make a solution that is 28% alcohol by volume?

- (A) 84 (B) 300
(C) 45 (D) 60

20.

HARD · NO CALCULATOR NEEDED

The number of tickets N remaining w weeks after sales began is modeled by $N = 1200 - 45w$. Consider the following statements.

- I. 1,200 tickets were available when sales began.
II. Each ticket sold reduces the number of remaining tickets by 45.
III. The model predicts 300 tickets remain after 20 weeks.

Which of the statements are true?

- (A) I only (B) I and II only
(C) I and III only (D) I, II, and III

21.

HARD · CALCULATOR

The length L , in centimeters, of a metal rod at temperature T degrees Celsius is modeled by $L = 120 + 0.0024T$. The temperature of the rod rises by 90 degrees Fahrenheit. By how many centimeters does the length of the rod increase?

- (A) 0.389 (B) 0.077
(C) 0.120 (D) 0.216

22.

HARD · NO CALCULATOR NEEDED

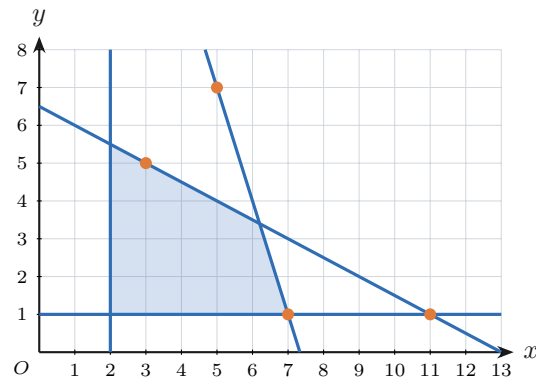
A triangle has side lengths x , $2x + 1$ and 12, where x is a positive integer. How many different values of x are possible?

- (A) 6 (B) 7
(C) 8 (D) 5

23.

HARD · NO CALCULATOR NEEDED

A bakery makes x loaves of rye and y loaves of sourdough each day. The shaded region shows all combinations (x, y) the bakery can produce, with the boundaries shown as solid lines. On a day when the bakery makes 4 loaves of rye, what is the greatest number of loaves of sourdough it can make?



24.

SAT CHALLENGE · CALCULATOR

Plan A costs \$40 per month plus \$3 per gigabyte (GB) of data. Plan B costs \$25 per month for the first 4 GB plus \$8 for each GB beyond 4 GB, and then an 8% fee is added to the monthly total. For whole-number usage $g \geq 4$ GB, what is the least value of g for which Plan B costs more than Plan A?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$y \geq 3x + a$$

$$2y < 6x - 8$$

In the system of inequalities above, a is a constant. For the least integer value of a for which the system has no solution, what is the value of a ?

26.

SAT CHALLENGE · DESMOS

In the xy -plane, the region defined by $x \geq 0$, $y \geq 0$, $x \leq 5$, $3x + 2y \leq 18$ and $x + 2y \leq 10$ is a polygon. What is the area of the polygon?

27.

SAT CHALLENGE • CALCULATOR

A bakery's daily cost is \$120 plus \$0.75 per loaf. On one day it sells 500 loaves: the first 100 at a price of p dollars each and the remaining loaves at 80% of that price. What is the least price p , in dollars, that is a whole number of cents and gives a profit of at least \$200 that day?

28.

SAT CHALLENGE • CALCULATOR

A jeweler melts together 12 kilograms of an alloy that is 25% silver, some amount of an alloy that is 60% silver, and some amount of pure copper (which contains no silver) to make 40 kilograms of an alloy that is 36% silver. How many kilograms of pure copper were used?

- (A) 14.4 (B) 19
(C) 28 (D) 9

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

A triangle has side lengths $2x$, $x + 6$ and $18 - x$, where x is an integer. What is the sum of all possible perimeters of the triangle?

30.

SAT CHALLENGE · DESMOS

Candle A is 24 centimeters tall and burns down at a constant rate of 2 centimeters per hour. Candle B is 30 centimeters tall and burns down at a constant rate of 3.5 centimeters per hour. Both are lit at the same time, and each stays at 0 centimeters once it has burned out. For how many whole numbers of hours t , with $t \geq 0$, is candle A at least 3 centimeters taller than candle B?

(A) 5

(B) 6

(C) 3

(D) 4

31.

SAT CHALLENGE · CALCULATOR

A utility charges a \$15 service fee each month. Usage is billed at \$2.40 per unit for the first 20 units and \$3.60 for each unit beyond 20. A 10% fuel surcharge is added to the usage charge only (not to the service fee). Then 5% tax is applied to the sum of the service fee and the surcharged usage charge. For $u > 20$ units, which function gives the total monthly bill $B(u)$, in dollars?

(A) $B(u) = 3.78u - 9.45$

(B) $B(u) = 4.158u - 11.97$

(C) $B(u) = 4.158u - 10.395$

(D) $B(u) = 4.158u - 12.72$

32.

SAT CHALLENGE · CALCULATOR

A utility computes a customer's monthly water bill B , in dollars, for u units of water in this order: a service fee of \$18 plus \$2.40 for each unit used; then a one-time credit of \$12 is subtracted; then a 5% surcharge is added to the amount at that point; and finally a fixed \$3.50 fee is added. Which of the following defines B for u units?

(A) $B = 2.52u + 9.975$

(B) $B = 2.52u + 22.4$

(C) $B = 2.52u + 9.8$

(D) $B = 2.52u + 10.4$

33.

SAT CHALLENGE · CALCULATOR

A ride service charges a fixed pickup fee plus a constant rate for each mile traveled. A 5-mile ride costs \$14.50 and a 12-mile ride costs \$29.20. The service then raises its per-mile rate by 20% and leaves the pickup fee unchanged. Using 1 mile = 1.6 kilometers, what will a 20-kilometer ride cost after the change?

- (A) \$35.50 (B) \$36.30
(C) \$54.40 (D) \$84.64

34.

SAT CHALLENGE · DESMOS

A coffee shop blends Ethiopian beans, which cost \$18 per kilogram, with Colombian beans, which cost \$11 per 500 grams. The blend has a total mass of 36 kilograms, and its cost works out to \$20.50 per kilogram. How many kilograms of Ethiopian beans are in the blend?

- (A) 22.5 (B) 18
(C) 27 (D) 13.5

35.

SAT CHALLENGE · CALCULATOR

Plan P costs \$25 per month for up to 4 gigabytes (GB) of data, plus \$6 for each additional GB. Plan Q costs \$14 per month for up to 2 GB of data, plus \$0.008 for each additional megabyte (MB). Charges for additional data are prorated, and 1 GB = 1000 MB. For what amount of data, in MB, do the two plans cost the same?

36.

SAT CHALLENGE · CALCULATOR

Plan A charges a one-time enrollment fee of \$85 plus \$18.50 per month. Plan B has no enrollment fee and charges \$28 per month. For a whole number of months m , which of the following describes all values of m for which Plan A costs strictly less than Plan B in total?

- (A) $m \geq 10$ (B) $m \geq 8$
(C) $m \geq 9$ (D) $m \leq 8$

37.

SAT CHALLENGE · DESMOS

$$x \geq 0, \quad y \geq 0, \quad 2x + y \leq 10, \quad x + 3y \leq 15, \quad 3x + 2y > k$$

In the system of inequalities above, k is a constant. What is the least value of k for which the system has no solution (x, y) ?
